

Cross-Platform Colon Cancer Proliferation Classification

Leakage-Free ML | StabilitySelector | Calibration Benchmark | Drug Screen

ScienceMontgomery 2025 | Ronisnotasianfr | Project ID: CS-XXX

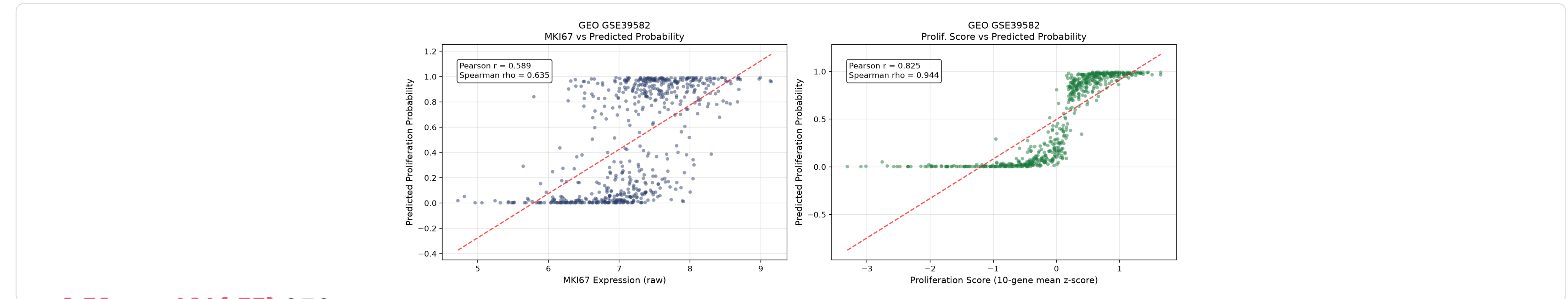
GEO (GSE39582 n=585, GSE17538 n=232) | TCGA-COAD (n=329) | CPTAC (n=105) | GDSC2

Problem & Motivation

Colon cancer is the 2nd leading cause of cancer death.
Ki-67 IHC is subjective, single-gene, no drug prediction.

Can ML predict proliferation from transcriptomics
without data leakage?*

Ki-67 Correlation (Leakage-Free)

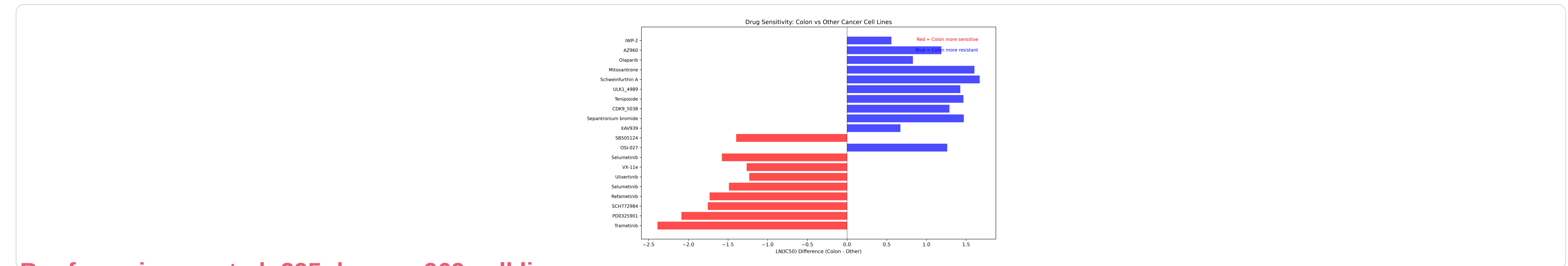


r = 0.59, p < 10⁻⁵⁵ GEO:

r = 0.54, p < 10⁻²⁵ TCGA-COAD:

10 proliferation genes removed from features; model
infers proliferation via downstream transcriptional cascades.

Drug Sensitivity (GDSC2)



Bonferroni-corrected: 295 drugs x 969 cell lines

Trametinib (MEK) p=1.8e-12 PD0325901 (MEK) p=5.9e-12
SCH772984 (ERK) p=1.1e-10 Refametinib (MEK) p=2.7e-10

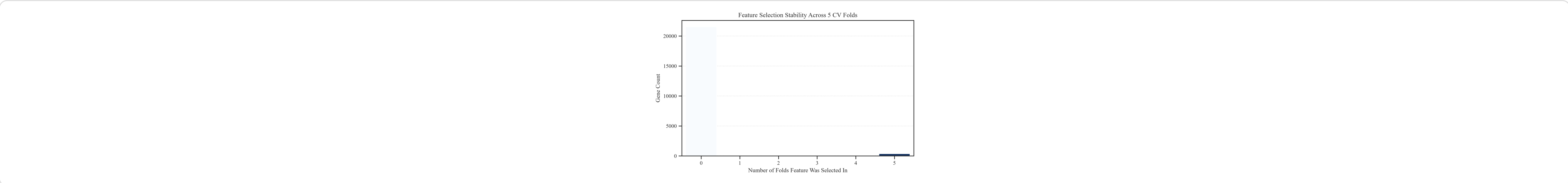
Top 5 = MAPK/ERK pathway (biologically validated)

Pipeline & Methodology

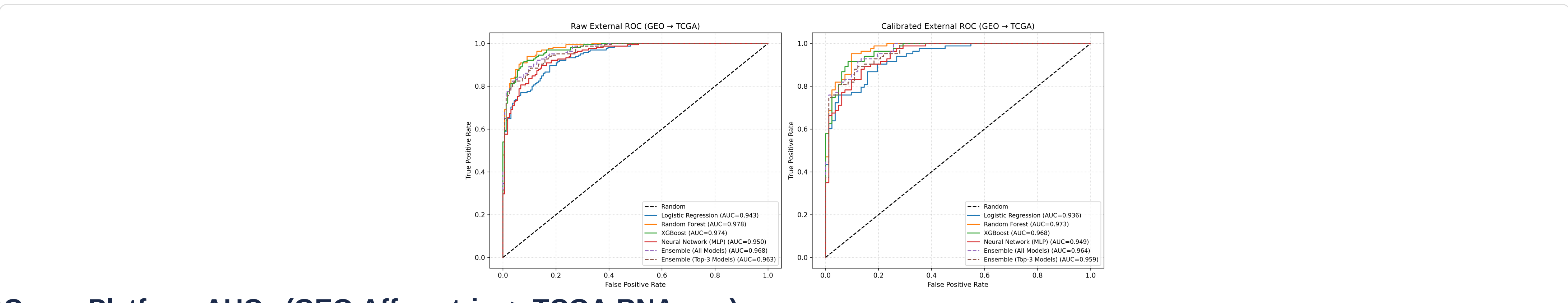
GEO GSE39582 (585) + GSE17538 (232) -> Train
Remove 10 proliferation genes (anti-leakage)
StabilitySelector: bootstrap 100x, keep >=50%
4 models: LR, RF, XGBoost, MLP (nested CV)
TCGA-COAD (329) + CPTAC (105) -> Validate
5 calibration methods (Platt, Isotonic, QN, ...)
GDSC2: 295 drugs x 969 lines (Bonferroni)

StabilitySelector: Novel sklearn Transformer

Bootstrap 100x | ANOVA F-scores per resample | Retain
features in top-K across >=50% of resamples



External Validation: GEO -> TCGA



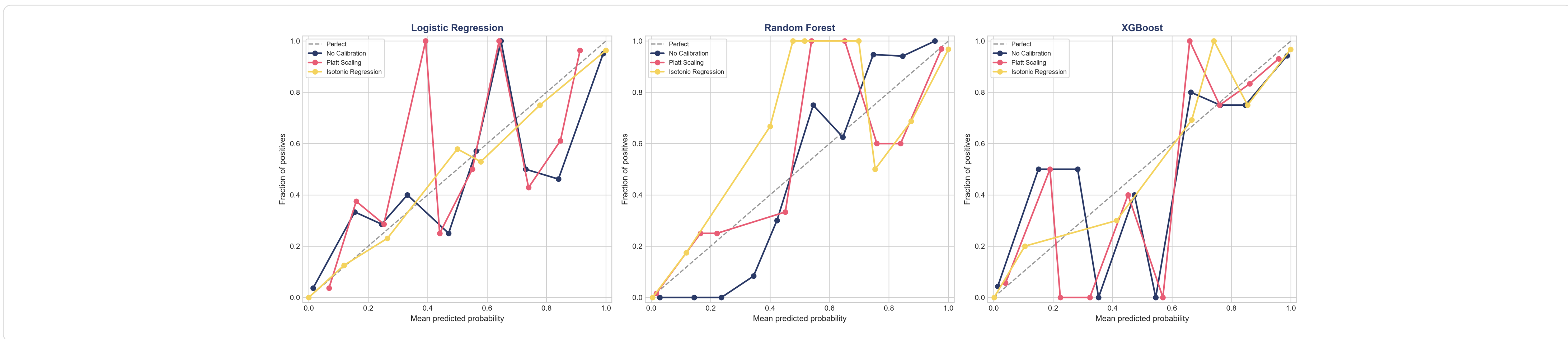
Cross-Platform AUCs (GEO Affymetrix -> TCGA RNA-seq)

Logistic Regression Raw: 0.974 Calib: 0.963
Random Forest Raw: 0.962 Calib: 0.952
XGBoost Raw: 0.971 Calib: 0.964
Neural Network (MLP) Raw: 0.975 Calib: 0.967
Ensemble (All Models) Raw: 0.976 Calib: 0.967
Ensemble (Top-3 Mode) Raw: 0.978 Calib: 0.969

Calibration Benchmark Results

Key: Platt Scaling Significantly Reduces ECE

RF Platt: AUC=0.973, ECE=0.043 — CIs non-overlapping with None
LR QN+Platt: AUC=0.972 — best cross-platform calibration



Best Config (sorted by ECE)

Logistic Regre Isotonic Regressio AUC=0.925 ECE=0.028
Neural Network Isotonic Regression AUC=0.935 ECE=0.029
Random Forest Platt Scaling AUC=0.973 ECE=0.043
XGBoost Platt Scaling AUC=0.968 ECE=0.038

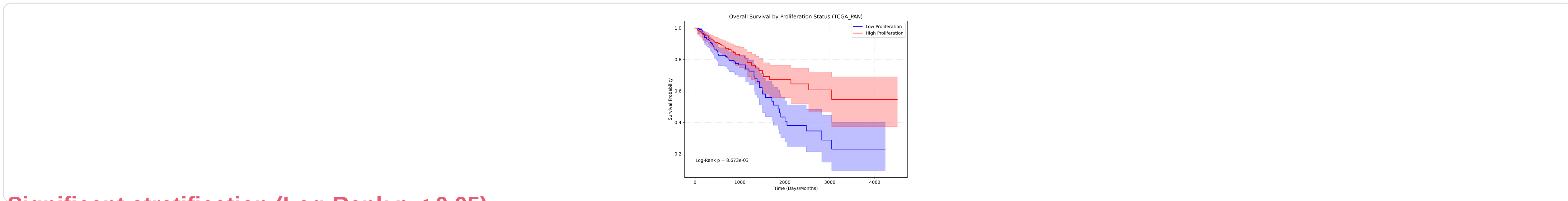
Ablation: StabilitySelector vs SelectKBest

SKB Logistic Reg Holdout AUC = 0.994
SS Logistic Reg Holdout AUC = 0.994
SKB Random Fores Holdout AUC = 0.988
SS Random Fores Holdout AUC = 0.988
SKB XGBoost Holdout AUC = 0.991
SS XGBoost Holdout AUC = 0.990
SKB Neural Netwo Holdout AUC = 0.983
SS Neural Netwo Holdout AUC = 0.992

SS matches or beats SKB on all 4 models

MLP: +0.009 AUC (0.983 -> 0.992) Others: equivalent

Survival Analysis



Significant stratification (Log-Rank p < 0.05)

GSE39582 (GEO) p = 0.037 (N=585)
TCGA-COAD p = 0.034 (N=329)
TCGA PanCancer p = 0.009 (N=434)